IT 112: Web Systems & Technologies

BSIT-2B 2nd Sem

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QuickReserve: Online Bus & Travel Booking System

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Introduction:

QuickReserve is a web-based application designed to simplify the process of booking bus tickets online. Instead of going to terminals or relying on manual systems, users can search for trips, select seats, and confirm bookings through a digital platform. The system improves convenience, reduces waiting time, and enhances the overall travel experience.

System Purpose:

The purpose of QuickReserve is to provide a fast, reliable, and user-friendly platform for booking bus tickets while allowing administrators to efficiently manage trips, schedules, and passenger data.

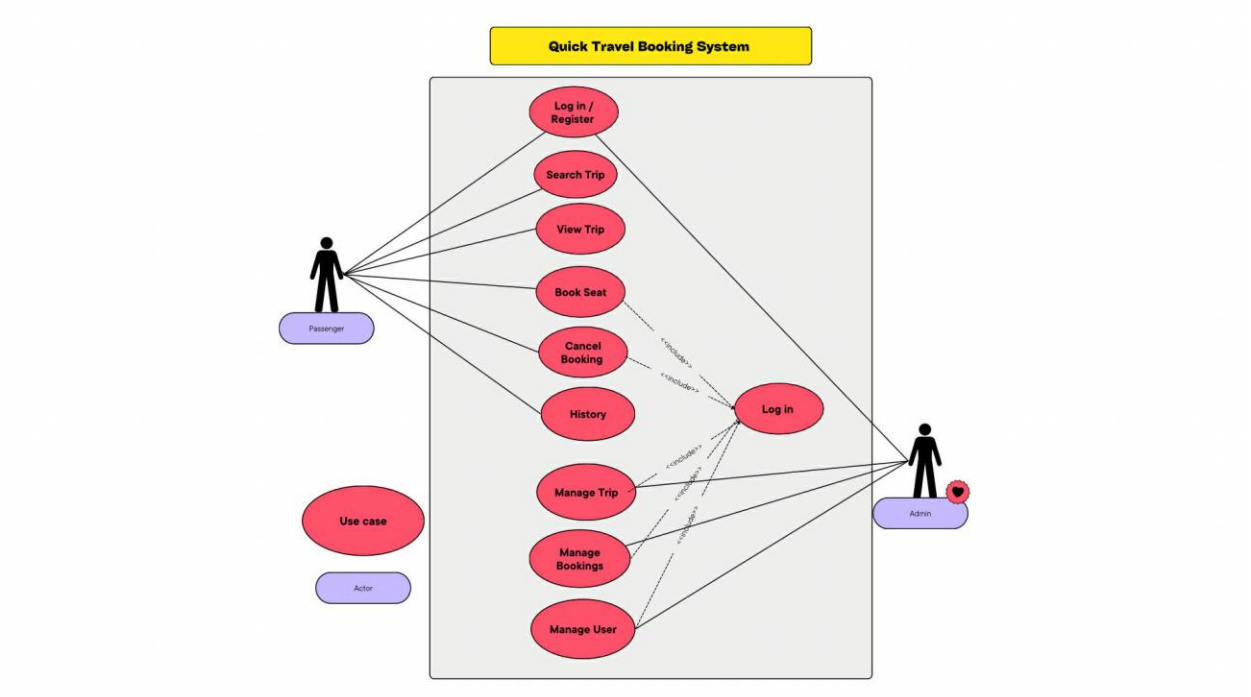
Target Users:

- **Passenger (User)** – Books trips and manages reservations
- **Admin (Operator)** – Manages trips, schedules, and bookings

Key Features:

- User registration and login
- Search trips (origin, destination, date)
- Seat selection and booking
- View and cancel bookings
- Admin management of trips and users

2. USE CASE DIAGRAM



Actors:

- Passenger
- Admin

Passenger Use Cases:

- Register / Login
- Search Trips
- View Available Trips
- Book Seat
- Cancel Booking
- View Booking History

Admin Use Cases:

- Add Trip
- Edit Trip
- Delete Trip

- Manage Bookings
- Manage Users

3. ERD (MongoDB STYLE)

User

_id: ObjectId
name: String
email: String
password: String
role: String
createdAt: Date

Trip

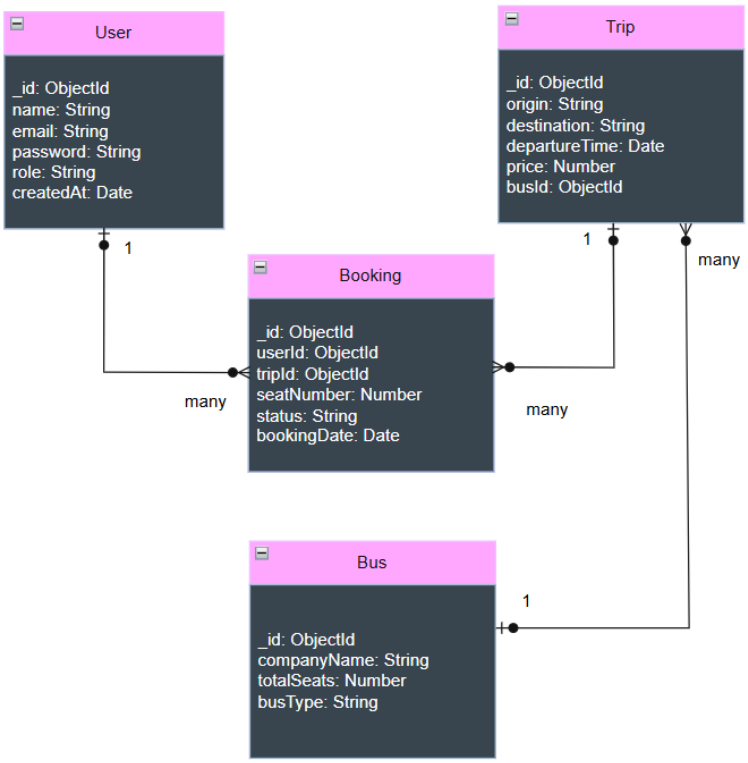
_id: ObjectId
origin: String
destination: String
departureTime: Date
price: Number
busId: ObjectId

Booking

_id: ObjectId
userId: ObjectId
tripId: ObjectId
seatNumber: Number
status: String
bookingDate: Date

Bus

_id: ObjectId
companyName: String
totalSeats: Number
busType: String

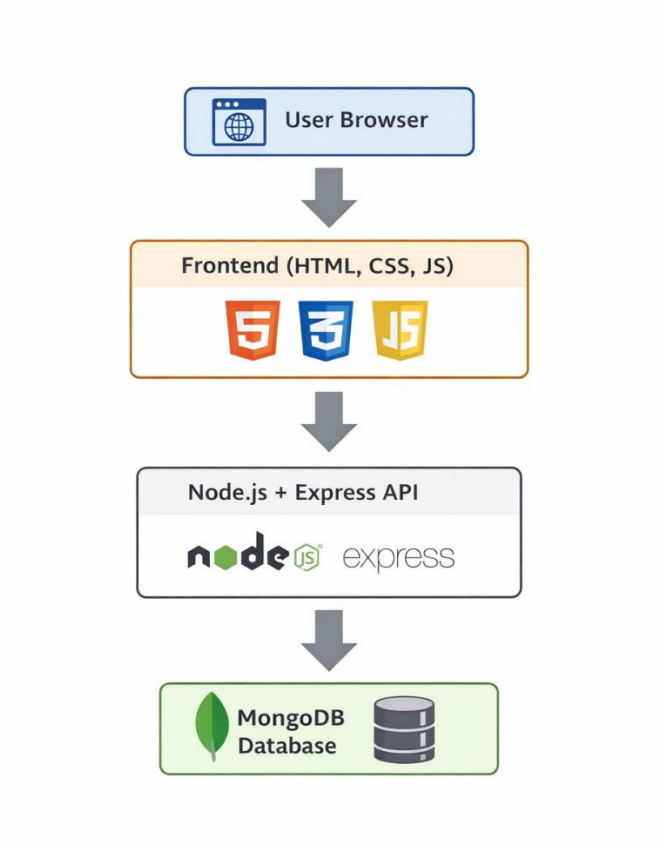


Relationships

User (1) - (many) Booking
Trip (1) - (many) Booking
Bus (1) - (many) Trip

4. SYSTEM ARCHITECTURE

Diagram Content:



Flow Explanation:

1. User searches for trips
2. Frontend sends request to backend
3. Backend retrieves data from MongoDB
4. Results are displayed to user
5. User books seat → saved in database